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Rice Fields Insurance Mobile Application

Final Project
in

IT-311

System Integration and Architecture 2

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Chapter 1

THE PROBLEM AND ITS SCOPE

1.1 Introduction

Rice farming is one of the main livelihood of most Filipino farmers and agriculture is important for the economy and food security of the country. But there are always dangers such as floods, droughts, pests, diseases, and other natural calamities that could lead to a large crop loss and financial suffering to rice producers. Mahul, O., & Stutley, C.J. (2010). Crop insurance systems are vital for stabilizing farmers' income and mitigating the financial impact of disasters.

The process of applying for rice field insurance in the Municipality of San Pablo, Zamboanga del Sur, is still mostly done manually. Farmers have to go to the Municipal Agriculture Office to submit forms and accompanying documentation and this often results in long waits, delays, incomplete submissions and increased travel costs. Such problems may dissuade farmers from patronizing insurance services, and cause delays in the processing of applications.

The study aims to design a Mobile Application for Rice Fields Insurance to tackle the above issues. The technology allows farmers to register, apply, upload papers and track their insurance status online. It strives to increase efficiency, accessibility and communication, while offering faster and clearer insurance services for the farmers of San Pablo.

1.2 Purpose and Description

The Rice Fields Insurance Mobile Application is an application that helps local farmers to protect their rice fields from natural catastrophes, pests, illnesses, and

other unforeseen calamities. Its main objective is to defend the livelihood of the farmers, maintain the productivity of the agricultural land, and contribute to the food security of the Municipality of San Pablo. The app also offers farmers a convenient and accessible platform to easily register, apply for insurance and upload required papers using their mobile devices. It also enables users to check the status of their application in real time and lessens the need to contact the Municipal Agriculture Office from time to time.

The initiative provides a digital platform where farmers can easily register, apply and monitor their insurance coverages, eliminating the need for manual paperwork and long delays at the Municipal Agriculture Office. The solution automates procedures such as farmer profiling, insurance application and approval, claims monitoring and data management for faster, more accurate and transparent services. The idea aims to secure farmers' income, enhance reliable rice production and create resistance to climate change and unforeseen dangers.

Objectives

This proposed system aims to:

- Design a system that's easy to use and can help farmers register apply for insurance and upload documents.

- Develop the Rice Fields Insurance Mobile Application that can automate the process of getting insurance for rice fields.

- Test the system to make sure it works well and can help farmers get insurance for their rice fields.

1.4 Project Context

Rice Fields Insurance Mobile Application is an application created to modernize and improve the current manual process for rice field insurance in the Municipality of San Pablo, Zamboanga del Sur. Today, farmers apply for insurance through traditional paper-based processes, which usually leads to long waits, delays, incomplete submissions and higher travel costs. These issues do not only affect the farmers but also the efficiency of the Municipal Agriculture Office's (MAO) in processing and handling insurance applications. To overcome these obstacles the suggested system provides a digital platform for farmers to register, apply for the insurance, upload essential paperwork and check the status of their application online. The technology enables the Municipal Agriculturist to assess and verify and approve applications efficiently and save accurate and ordered information in a centralized database.

The study agrees with the increasing application of information and communication technologies in the agricultural industry to better service delivery, transparency and accessibility. As the insurance process becomes digital, the technology reduces the amount of manual work, the number of human errors and the processing time for applications and claims. This initiative contributes to the greater objective of improving agricultural resilience by offering farmers easy access to insurance services to protect them from financial losses due to natural disasters, pests and other threats. This innovation also helps to the improvement of agricultural output, greater farmer support services, and sustainable development in the community under the San Pablo Rice Fields Insurance System.

1.5 Scope and Limitations

The Rice Fields Insurance Mobile Application is a system created to support the farmers in the Municipality of San Pablo Zamboanga del Sur. The technology can assist farmers in registering, applying for insurance and uploading paperwork. But there are certain limits with the Rice Fields Insurance Mobile Application. The application is only for rice fields in the Municipality of San Pablo Zamboanga del Sur.

They may register, apply for insurance and upload papers quickly with this app. But there are certain limits in utilizing the Rice Fields Insurance Mobile App. It only works for rice fields at San Pablo Zamboanga del Sur. Farmers need a device with internet to utilize and get insurance for their rice fields . And also means farmers need to have an internet connection and a gadget that can connect to the internet.

1.6 Definition of Terms

Rice Fields Insurance Mobile Application. is a way to assist farmers to receive insurance for their rice fields.

Municipal Agriculturist. the person in charge of the Municipal Agriculture Office.

Farmer. is a person who cultivates rice and uses the Rice Fields Insurance Mobile Application.

Chapter 2

REVIEW OF RELATED STUDIES AND SYSTEM

2.1 Foreign

“ The Effect of Rice Farm Insurance on the Income of Farmers in Indonesia. Agricultural,”

According to Harmini et al. (2021) “The Impact of Rice Farm Insurance on the Income of Farmers in Indonesia. “Agricultural” insurance is necessary for farmers to protect them from significant losses and for them to have working capital for the next farm.

This is related to the study since the major objective is also to protect the farmers’ livelihood through the insurance of rice fields from natural catastrophes, pests and diseases, providing financial protection and stable agricultural productivity in the municipality.

“Farmers' attitudes towards crop insurance and their readiness to pay in the Çarşamba area of Samsun, Turkey. International Journal of Development and Sustainability”

Attitude of farmers toward agricultural insurance and willingness to pay in Çarşamba district of Samsun, Turkey. Kipkemoi, J. K. and Ceyhan, V. (2021). Studies in International Journal of Development and Sustainability’ show that farmer participation in insurance programs indicates an awareness that insurance acts as a tool for mitigating risks arising from uncertainties in farming, although variables such as information access, cost and trust can affect adoption choices.

This is relevant to the study as it allows for greater farmer enrollment through a simpler digital platform that reduces cost, increases access to information and trust in the insurance process.

“Enhancing Insurance Risk Management in the Agricultural, Livestock and Fishery Sectors of Thailand”

As per the study of Chaiyawat, T., Mekkriengkrai, S., & Phanwichit, S (2023) “Enhancing the Effectiveness of Insurance Risk Management in the Agricultural, Livestock, and Fishery Sectors in Thailand,” the governmental support and financial subsidies are the appropriate and effective methods in the management of agricultural risks. These actions contribute to making crop insurance more attractive and financially viable for farmers, thus improving its overall effectiveness.

This is related to the study since the major goal of the study is also to preserve the livelihood of the farmers by insuring rice fields against natural catastrophes, pests and illnesses, giving financial protection and stable agricultural productivity in the municipality.

2.2 Locality

“Implementation of the Philippine Crop Insurance Corporation (PCIC) Registry System for the Basic Sector in Agriculture (RSBSA)”

“Philippine Crop Insurance Corporation (PCIC) Registry System Implementation in the study of Philippine Crop Insurance Corporation (2021) The “Basic Sector in Agriculture (RSBSA)” in the report on delays and backlogs in the insurance procedure raises worries about long-term satisfaction. The first reactions were positive, but Himmelfarb

(2023) and Reario (2024) cautioned that such satisfaction might be short-lived if complex paper-based processes and slow service delivery continue.

This is highly related to the research as the manual, paper-based procedure in the municipality creates hurdles in all the three categories. Limited access to clear information for farmers, significant indirect costs from travel and delays and loss of confidence in the system owing to inefficiency. The San Pablo system addresses these hurdles by digitizing applications, permitting document uploads, and providing real-time status updates, making it easier, cheaper, and more trustworthy to participate.

"Three Key Reform Areas to Improve Philippine Agricultural Insurance to Help Farmers Deal with Natural Disasters"

World Bank (2024). "Transforming agricultural insurance in the Philippines to support farmers to cope with natural disasters: Three key reform areas". Noted the persistent operational challenges including inaccessibility to farms, poor communication channels and limited adjuster capacity, which affect timely claim resolution and damage the credibility of agricultural insurance schemes.

Critical ways to increase efficiency and recover farmer trust include modernizing procedures and growing the adjuster personnel, according to the research. This is associated with the study that solves these problems with the aid of a digital platform that provides quicker and more transparent services.

"Philippines Satellite-Based Area-Based Yield Index (ARBY) Crop Insurance Pilot (2025)

(2025) Satellite-Based Area-Based Yield Index (ARBY) Crop Insurance Pilot in the Philippines PCIC et al., (2025) In June 2025, the Philippine Crop Insurance Corporation (PCIC), Philippine Rice Research Institute (DA-PhilRice), PAGASA, the International

Rice Research Institute (IRRI) and CIAT launched a pilot program deploying Area-Based Yield Index Insurance (ARBY) with weather-informed agro-advisory services. Payouts based on the yield index are based on past rice yields from the Philippine Rice Information System (PRiSM) under the new crop insurance model that eliminates the need for on-site damage inspections and makes the insurance procedure faster. It covers hazards such as floods, drought, saltwater intrusion, pests and diseases and is bundled with the Climate+ tool to provide timely weather-based field advisories to farmers.

This was linked to the proposed Rice Fields Insurance Mobile Application because both are intended to enhance the efficiency, accessibility and transparency of agricultural insurance services for farmers. The ARBY program uses satellite technology, historical yield data and weather-based advisories to make it easier to assess insurance and to speed up the process of settling claims. Likewise, the Rice Fields Insurance Mobile Application digitizes the insurance process by enabling farmers to register, apply, upload requirements, and monitor the status of their insurance through their mobile devices.

Chapter 3

DESIGN AND METHODOLOGY

3.1 4D Model

The study used the Research and Development (R&D) process applying the 4D Model (Define, Design, Develop, and Deploy) in constructing the suggested system.

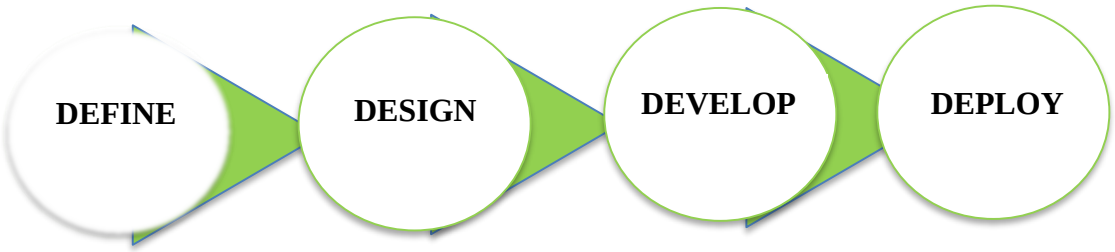


Figure 1- 4D Model

This figure depicts the 4D Model (Define, Design, Develop, and Deploy), which acts as the basis for the creation of the San Pablo Rice Fields Insurance. The approach was chosen because of its simple yet effective process in producing working systems. It gives a step-by-step method to verify that the proposed insurance system is clearly defined, appropriately designed, extensively built, and efficiently deployed to its intended users.







Image Interview:

The photographs show the researchers doing a data-gathering activity through an interview in an office setting. The process begins when the researchers visit the office and approach the staff to request permission to conduct their study. They introduce themselves and describe the goal of their research. After getting approval, the researchers proceed with the interview. They ask the personnel crucial questions and collect important information relating to their study. They also examine the office environment and the process that is being followed and jot down the replies throughout this stage. During the interview, the researchers explain the answers provided by the staff, to ensure the accuracy and completeness of the data. They will analyze the information collected carefully and may ask follow up questions if needed. The researchers are able to successfully do the interview procedure and data gathering overall that will aid to support and reinforce their research study.

The **define phase** The define phase identifies the problems in the manual rice field insurance procedure. The problems include long waits, delays, incomplete submissions, and high expenses of travel for farmers . It also involves the collection of farmer's and Municipal Agriculture Office's requirements.

During the **design phase** we will develop the architecture and operation of the digital system. This will include elements such as farmer registration, online application forms and document uploads. The emphasis is on user-friendly interfaces so that even farmers with no computer abilities can access the information.

The **develop phase** covers actual coding and development of the web-based system with core capabilities such as profiling farmer, application monitoring, approval

procedures. This stage is for evaluating each feature to assure the system's accuracy and reliability.

The **deploy phase** is about testing and demonstrating the San Pablo Rice Fields Insurance in a controlled environment. Instead of instant implementation of the system in the whole municipality, the system was first tested in sample testing and assessment inside the Municipal Agriculture Office of San Pablo. Thus, the researcher and the Municipal Agriculturist were able to see the system in action as the farmers registered, papers were uploaded, and application status was tracked. The demonstration helped to analyze the correctness of data processing, the responsiveness of system functionalities and the general reliability of the platform. With this pilot deployment, the researcher could test that the system can deliver faster, more transparent and organized insurance processing before being fully implemented in the municipality.

3.2 Current System Flow

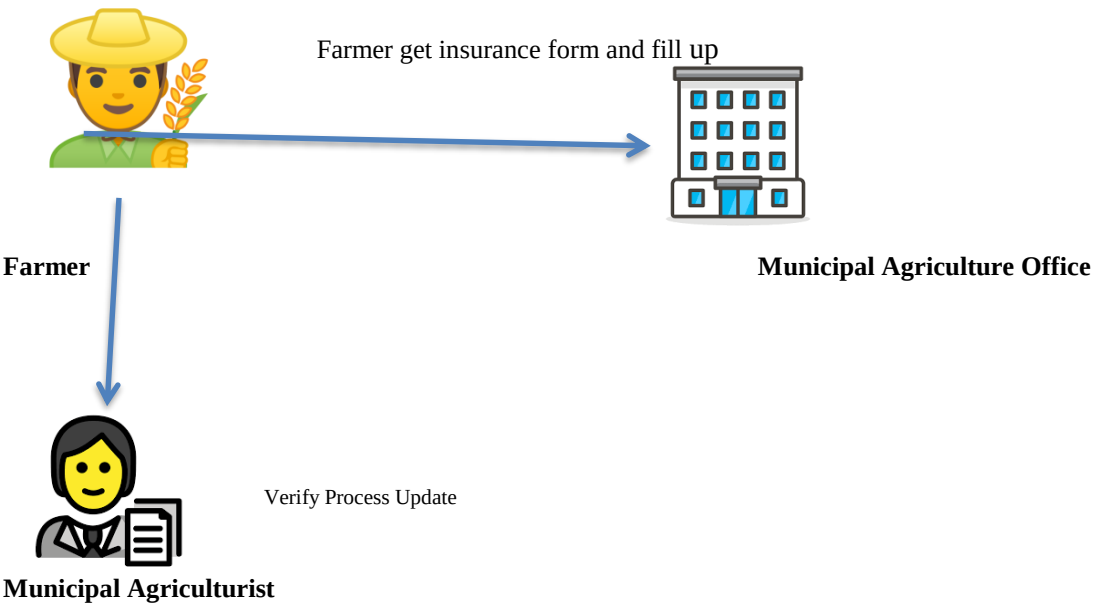


Figure 2- Current System Flow

The figure above shows the current process followed by farmers when applying for rice field insurance in the Municipality of San Pablo, Zamboanga del Sur. At present, farmers manually obtain and fill out insurance forms from the Municipal Agriculture Office (MAO). After completing the form, they personally submit it to the Municipal Agriculturist for verification. The Agriculturist checks the information and waits for updates or confirmation from the Philippine Crop Insurance Corporation (PCIC) regarding the farmer's eligibility and indemnity. Once the verification and approval from PCIC are received, the Municipal Agriculturist posts an updated list of farmers who are insured and qualified to claim benefits. However, because the entire process is done manually, it often causes delays, long queues, and incomplete submissions. Farmers also need to travel to the office to file or follow up on their applications, which results in wasted time and effort. This current manual process shows the need for a more efficient, transparent, and accessible digital system that can automate application submission, verification, and update posting.

3.3 Context Diagram

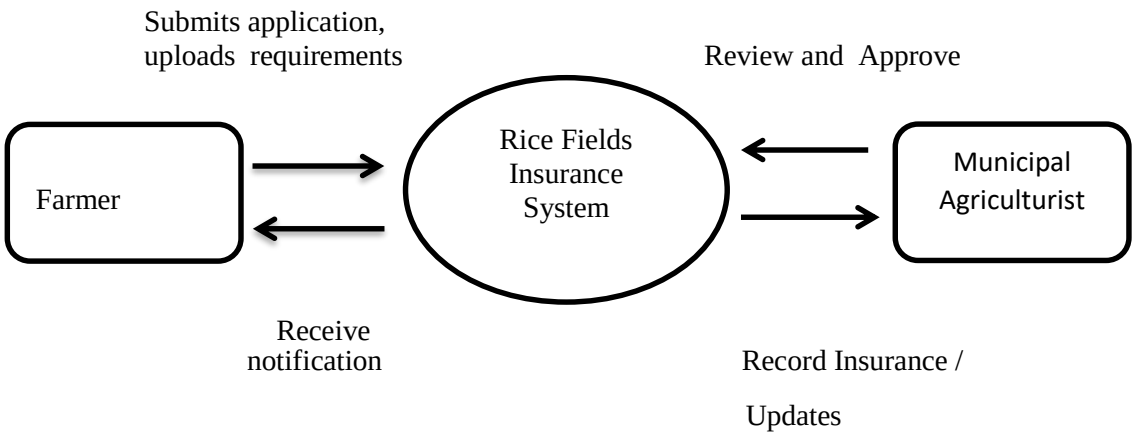


Figure 3- Context Diagram

The Context Diagram of the San Pablo Rice Fields Insurance System shows the interaction of the system and its external actors, the Farmer and the Municipal Agriculturist. The system is the main procedure for linking users to manage and process rice field insurance applications. Farmers can apply for insurance, provide required papers and receive updates or notifications on the status of their application. The Municipal Agriculturist shall verify the supplied information, review and approve legitimate applications and record the authorized insurance details in the system database. San Pablo Rice Fields Insurance System handles these transactions by recording, validating and storing the data in real time. The interaction ensures an organized and effective flow of information between the farmers and the Municipal Agriculturist which will help to make decisions faster and reduce manual errors or delays. The method also fosters transparency, precision and accountability in processing of crop insurance for the farmers in San Pablo, Zamboanga del Sur.

3.4 Activity Diagram

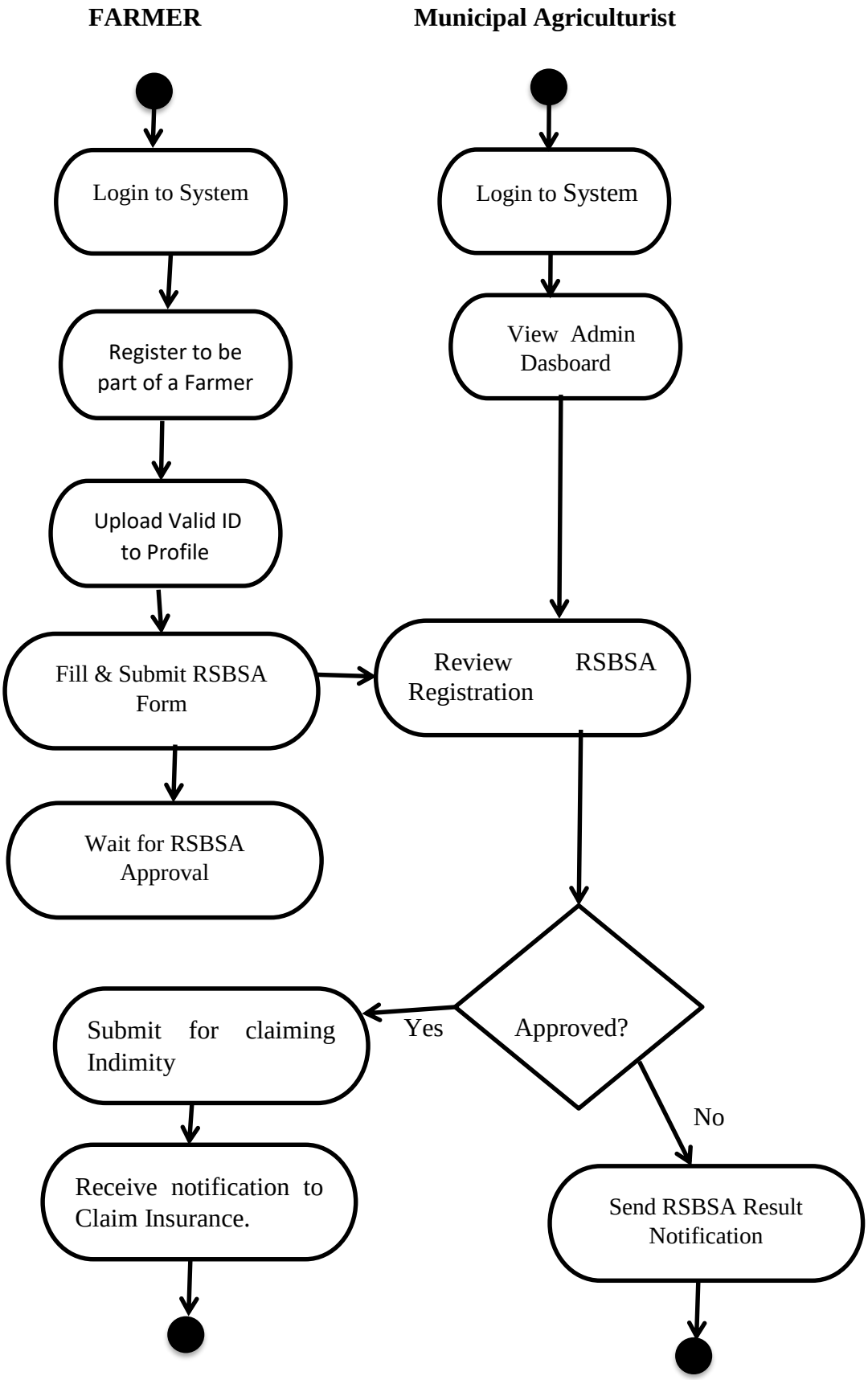


Figure 4 – Activity Diagram

The Activity Diagram shows how users interact with the system functions. The two main users in the system are the Farmer and the Municipal Agriculturist. The farmer would be able to execute tasks such as register an account, login into the system etc. using the system. The farmer can manage their profile and requirements by immediately uploading their ID, uploading supporting documents, and submitting their RSBSA registration through the program itself. These requests are maintained, verified and processed through the system by the Municipal Agriculturist. The admin login to the system to examine the submitted RSBSA registrations, check the validity of the supporting papers, and change the application statuses in the database.

3.5 USE Case Diagram

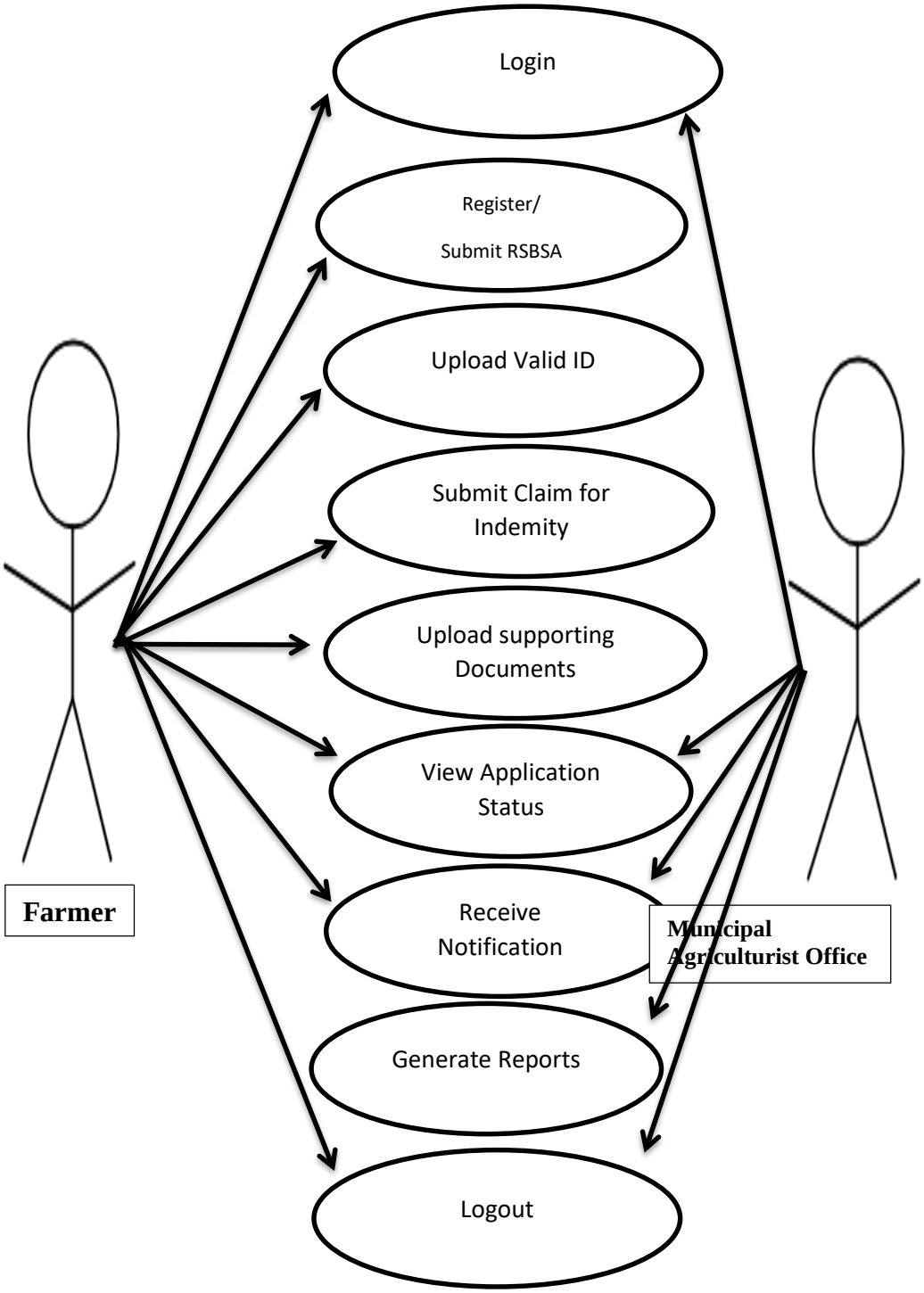


Figure 5 – USE Case Diagram

The Use Case Diagram displays the interaction between the users and the functionalities of the system. The two major users in the system are the farmer and the municipal agriculturist. The method for the rice insurance is meant to benefit the Farmer and the Municipal Agriculturist. The Farmer is using the system to perform things like: Log into an account, Register, Send in insurance applications, Upload paperwork, See the status of the application, Receive notification . This way the Farmer can accomplish things online instead of visiting to the Municipal Agriculture Office all the time. The Municipal Agriculturist shall examine the insurance applications. Also logs and look at the applications and paperwork examines the Farmers information register and apply to the applications puts down insurance details. Updates the statement of the result of the applications. The Municipal Agriculturist is also responsible for ensuring that all records are correctly maintained in the system.

REFERENCES

Foreign Studies

Harmini, H. (2022). *The Impact of Rice Farm Insurance on the Income of Farmers in Indonesia*. *Jurnal Manajemen & Agribisnis*, *19*(1), 59-68Liu, Z., Moon, I., & Zhang, R. (2024). Flexible vehicle scheduling with precedence constraints for tourists. International Transactions in Operational Research, 31(4), 2309-2337.

Kipkemoi, B. K., & Ceyhan, V. (2021). *Eliciting attitude of farmers toward agricultural insurance and willingness to pay in Çarşamba district of Samsun, Turkey.* *International Journal of Development and Sustainability*, *10*(4), 160–181.

Chaiyawat, T., Mekkriengkrai, S., & Phanwichit, S. (2023*). Enhancing the effectiveness of insurance risk management in agricultural, livestock, and fishery sectors in Thailand. Procedia of Multidisciplinary Research, 1*(7)

Local Studies

Philippine Crop Insurance Corporation. (2021). *Implementation of the Registry System for Basic Sectors in Agriculture (RSBSA) and its integration in agricultural insurance programs*. PCIC Annual Report 2021.

World Bank. (2024, April 12). *Transforming agricultural insurance in the Philippines to help farmers cope with natural disasters: Three essential reform areas*. World Bank Blogs.

PCIC, DA-PhilRice, PAGASA, IRRI, & CIAT. (2025). *Satellite-Based Area-Based Yield Index (ARBY) crop insurance pilot in the Philippines.* Philippine Crop Insurance Corporation, Quezon City.